

| Question Number | Answer<br>Additional guidance                                                                                                                                                                                                                                                                                               | Mark       |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| <b>7(a)(i)</b>  | <p>The only correct answer is B one</p> <p><i>A is not correct because the first statement is correct</i></p> <p><i>C is not correct because the generative nucleus divides to form two haploid male gametes</i></p> <p><i>D is not correct because the generative nucleus divides to form two haploid male gametes</i></p> | <b>(1)</b> |

| Question Number | Answer                                                                                                                                | Additional guidance                                                                             | Mark       |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|------------|
| <b>7(a)(ii)</b> | <ul style="list-style-type: none"> <li>• surface area calculated (1)</li> <li>• difference in standard form with units (1)</li> </ul> | <p>4.128 (<math>\mu\text{m}^2</math>)</p> <p><math>9.02 \times 10^{-1} \mu\text{m}^2</math></p> | <b>(2)</b> |

| Question Number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                    | Additional guidance                                                                                                                                                                                                                                                                                                    | Mark       |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| <b>7(b)</b>     | <p>A description that makes reference to the following points:</p> <ul style="list-style-type: none"> <li>• differential gene expression (1)</li> <li>• by {epigenetic modification / histone modification / DNA methylation} (1)</li> <li>• {proteins / enzymes} synthesised (from active genes) which (permanently) modify the cell (1)</li> <li>• description of modification to become a sclerenchyma cell</li> </ul> | <p>ACCEPT only some genes are {active / switched on} / some genes are switched off</p> <p>ACCEPT {proteins / enzymes} {made / synthesised} produce a {structural / functional / metabolic} change</p> <p>e.g. synthesis of {cellulose/ microfibrils / lignin} / secondary thickening / lignification of cell walls</p> | <b>(4)</b> |

| Question Number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Additional guidance                                                                                                                                                                                                                                                   | Mark       |
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| <b>7(c)</b>     | <p>An answer that makes reference to four of the following points:</p> <ul style="list-style-type: none"> <li>• meiosis increases genetic variation / crossing over and {random / independent} assortment occur in meiosis (1)</li> <li>• the {genes / alleles / loci} for colour and grain length are linked (1)</li> <li>• (therefore alleles for these traits) will be inherited together / (therefore alleles for these traits) unlikely to be separated during crossing over (1)</li> <li>• the {genes / alleles / loci} for pea pod wrinkles and {colour/grain length} are not linked (1)</li> <li>• (therefore) the alleles for these traits will be inherited independently due to independent assortment (1)</li> </ul> | <p>ACCEPT meiosis results in haploid cells</p> <p>ACCEPT linkage is when {genes / alleles} are close together on the same chromosome</p> <p>ACCEPT allele for this trait may be separated from {colour/grain length/ other alleles} due to independent assortment</p> | <b>(4)</b> |